

Two sample

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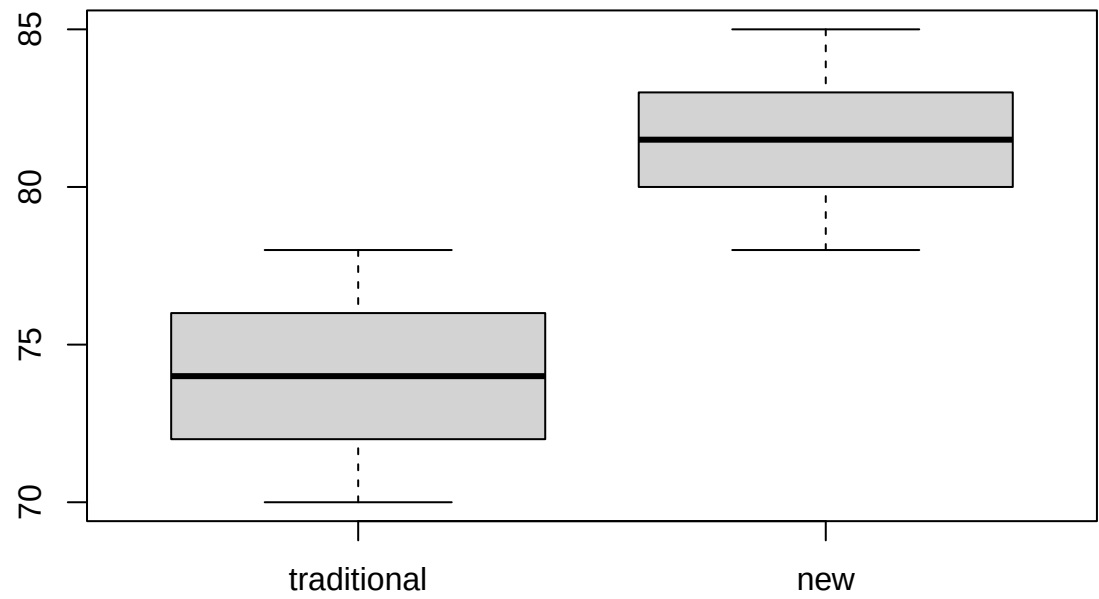
A study was conducted in a school to determine whether the new new method would improve student performance compared to the traditional method. Two samples of students were randomly selected and divided into two groups: one group was taught using the new method, and the other group using the old method.

```
##      traditional new
## 1           72  80
## 2           75  82
## 3           78  78
## 4           70  85
## 5           74  79
## 6           77  81
## 7           73  84
## 8           76  83
## 9           71  80
## 10          74  82
```

Table one : The table shows the distribution of student marks in both groups and indicates that the two groups are the same size. ‘ Descriptive statistic :

```
##          vars  n mean   sd median trimmed  mad min max range skew kurtosis
## traditional    1 10 74.0 2.58   74.0   74.00 2.97  70  78     8 0.00    -1.41
## new            2 10 81.4 2.22   81.5   81.38 2.22  78  85     7 0.09    -1.38
##              se
## traditional 0.82
## new         0.70
```

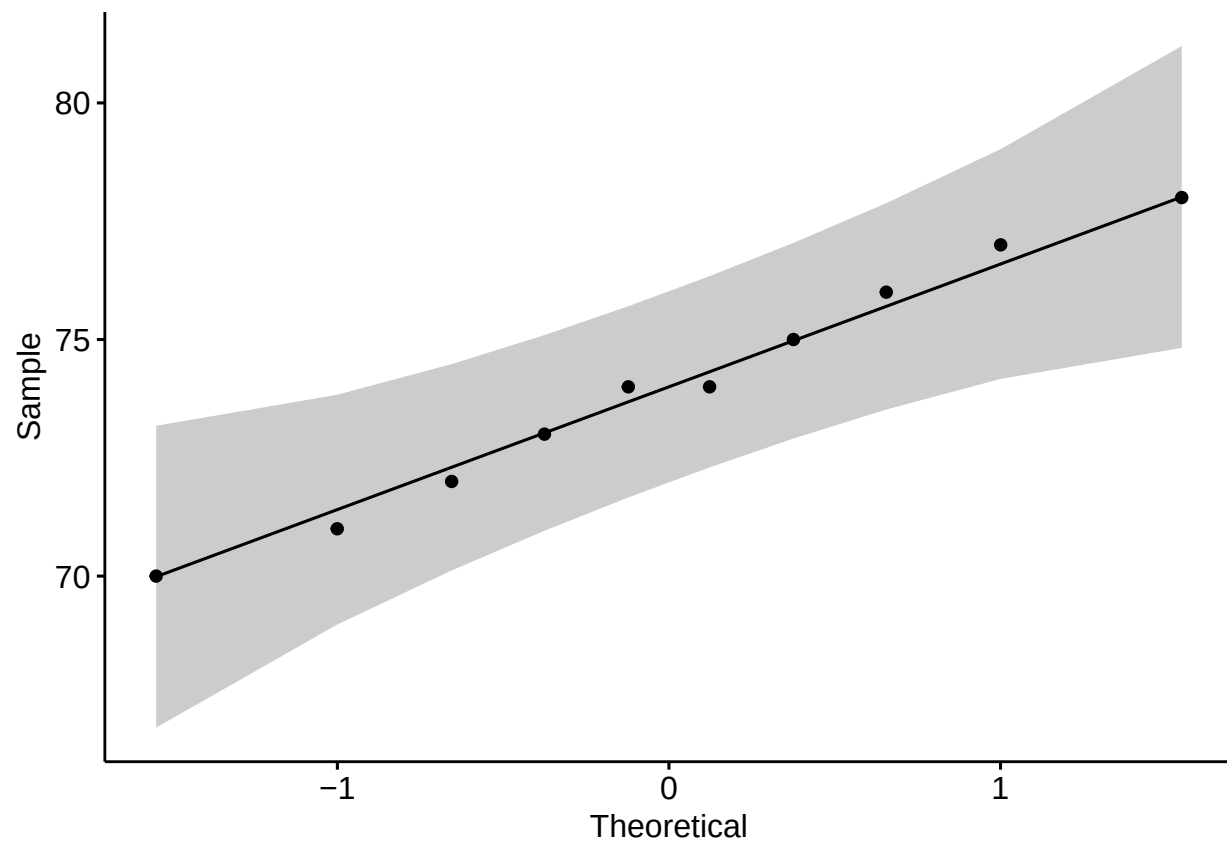
Table two : The results showed that the average is 74 and standarad deviation is 2.58 and the mean of new method is 81.4 and standard deviation is 2.22 The arithmetic mean for the new method was higher than that of the traditional method, meaning that students using the traditional method achieved less scores than students using the new method.

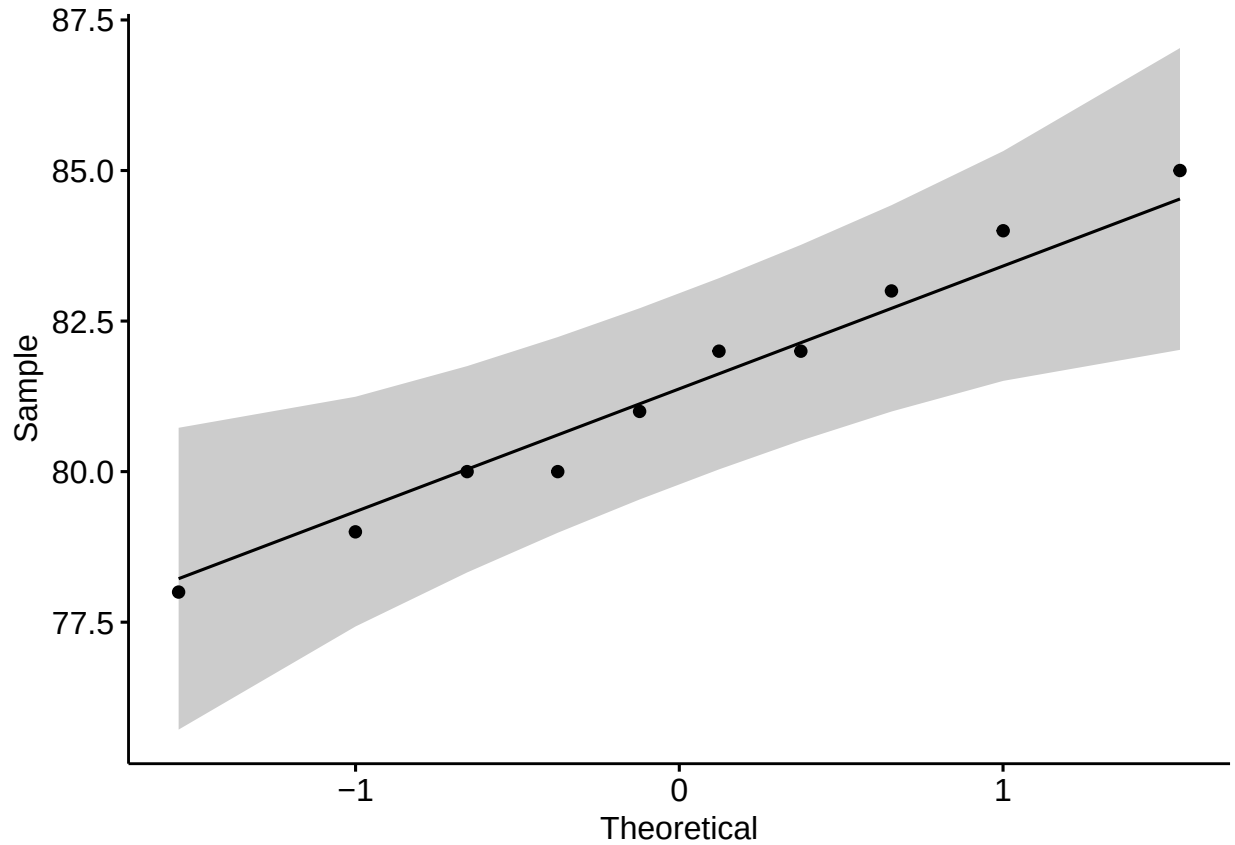


Graphical description :

figur one The graph shows that the new teaching method is better than the traditional, as the new group scored higher than the traditional group.

Inference statistics:





The data were verified to follow normal distribution using a gg-qq plot. The results showed that both groups follow normal distribution.

Homogeneity test:

```
## [1] 6.666667
```

```
## [1] 4.933333
```

```
## [1] 1.351351
```

The results showed that the two groups had the same homogeneity.

```
##
## Two Sample t-test
##
## data: traditional and new
## t = -6.8707, df = 18, p-value = 9.96e-07
## alternative hypothesis: true difference in means is less than 0
## 95 percent confidence interval:
##      -Inf -5.532356
## sample estimates:
## mean of x mean of y
##      74.0      81.4
```

The independent samples two-test was used to study the difference between the two methods. The results showed that there is a difference between the two methods, where the value of p-value = 9.96×10^{-7} less than 0.05 This means that the new method is better than the old one, This indicates that the new method improved student performance.